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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Bombaklat
Project / Solution Name	SKILLAX AI Platform
Name (Leader)	Cecilia Se
Email & Phone (Leader)	ceciliase1231@gmail.com
University Affiliation(s)	Universiti Sains Malaysia
Live Prototype URL	https://skillaxai.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/ado0330/SDC-Skillax

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The browser UI uses Next.js App Router and React state. Workspace state, including the profile and roadmap progress, is serialized to browser localStorage; no database is configured. The client sends CV text or a PDF to the Next.js *POST /api/cv/analyze* route. If a server-side Gemini key is configured, the route calls Google Gemini through *@google/genai* using the model ID in *GEMINI_MODEL* (default *gemini-3.5-flash-lite*); otherwise, or after a caught Gemini error, the code uses a keyword extractor. The structured proposal is returned for user review. The route sends content to Google when Gemini is active; provider retention/use depends on the account's billing tier and current terms. Career scores, gaps and roadmaps are calculated by local deterministic TypeScript functions. *GET /api/research* fetches Hugging Face and MIT RSS feeds and falls back to bundled signals. Workforce observations are synthetic defaults or user-imported data. *.replit* configures deployment commands, but a hosted instance is not verified.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Primary application programming language	Apache-2.0
Next.js 14.2.35	Full-stack React framework (App Router, API routes)	MIT
React / React DOM 18.3.1	UI component rendering engine	MIT
Tailwind CSS 3.4.19	Utility-first CSS styling framework	MIT
@google/genai 2.31.0	Official SDK for Google Gemini API integration	Apache-2.0

Name & Version	Purpose in Project	License
Zod 3.25.76	TypeScript-first schema validation for API responses & state	MIT
lucide-react 0.453.0	Icon set components for UI	ISC
clsx 2.1.1	Utility for constructing className strings conditionally	MIT
tailwind-merge 2.6.1	Utility to merge Tailwind CSS classes without style conflicts	MIT
PostCSS / Autoprefixer 8.5.31 / 10.6.1	CSS transformation and vendor prefixing tooling	MIT / MIT
tsx 4.23.15	TypeScript execute CLI tool for scripting	MIT
@types/node, @types/react, @types/react-dom 20.19.43 / 18.3.31 / 18.3.7	TypeScript type definitions for Node.js and React	MIT / MIT / MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (Google)	CV text analysis and structured proposal generation via @google/genai SDK (gemini-3.5-flash-lite)	Free Tier / Developer API Key
Hugging Face Blog RSS	Fetch research updates and machine learning blog signals for research endpoint	Free / Public RSS
MIT AI News RSS	Fetch artificial intelligence news feed for research endpoint	Free / Public RSS
Crossref Works API	For scholarly records matching health data, industrial IoT, project management, data analytics, and related topics.	Free / Public RSS

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
GitHub	Git remote points to https://github.com/ado0330/SCD-prototype2; configured source-control origin	Team to verify visibility/access

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts the SKILLAX web application and provides a live link for judges to access the prototype.	Standard Free Tier; deployment region: Asia

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	**Source / Provider**	**Usage (pre-trained / fine-tuned / self-trained)**	**License / Terms**
gemini-3.5-flash-lite	Google	Pre-trained (via @google/genai SDK)	Google API Terms of Service

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	**Source / URL**	**Type (public / private / self-collected / synthetic)**	**License**	**Personal Data? (Y/N)**
Hugging Face Blog RSS	https://huggingface.co/blog/feed.xml	Public	Public RSS Feed	N
MIT AI News RSS	https://news.mit.edu/rss/topic/artificial-intelligence2	Public	Public RSS Feed	N
Crossref scholarly metadata	https://api.crossref.org/works	Public	Public RSS Feed	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Antigravity	Developing the application's underlying logic, core functionality, and implementation workflow	Major
OpenAI Codex	Developing and refining the user interface (UI)	Major
ChatGPT	Seeking feedback, evaluating ideas, troubleshooting issues, and identifying improvements to the project	Moderate
Replit	Deploying the website and obtaining feedback and suggestions for further improvements	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team used Antigravity to develop the core application logic and functionality, and OpenAI Codex to build and refine the user interface. ChatGPT was used to seek feedback, evaluate design and implementation decisions, and identify potential improvements. Replit was used to deploy the website and obtain additional feedback for further refinement. AI-generated outputs were treated as development assistance rather than automatically accepted as correct. The team remains responsible for reviewing the implementation,

checking that the application behaves as intended, and addressing issues identified during testing and feedback.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core access / platform	Used to deploy the SKILLAX website and obtain feedback and suggestions to improve the prototype.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supported website deployment and iterative improvement of the SKILLAX prototype. It was useful for making the application accessible for demonstration and gathering feedback, but it was not essential to the core career-matching or skills-analysis logic. The application could be deployed on an alternative hosting platform, subject to compatibility and configuration requirements.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application operates as a client-side web workspace without a configured database backend. User profiles, selected career goals, and roadmap state are serialized directly to the browser's local storage and remain local to the user's device. When a user submits a CV (as raw text or PDF), the document text is processed server-side by the Next.js API route (/api/cv/analyze). If configured, the route transmits the CV text to the Google Gemini API for structured extraction; otherwise, a local fallback extractor processes the text. Data sent to the Gemini API is subject to Google's API data retention and privacy policies. No persistent user accounts, logs, or personal data are retained on application servers.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. Generative AI Dependency & Reliability: CV extraction relies on Google Gemini (gemini-3.5-flash-lite). Model outputs may occasionally produce incomplete or inaccurate skill extractions. If the API key is missing, or if rate limits/errors occur, the system falls back to a deterministic keyword extractor, which provides lower analytical depth. 2. Heuristic & Local Calculations: Career readiness scores, skill gap identification, and learning roadmaps are generated via local TypeScript heuristic algorithms rather than real-time labor market models. 3. Static & External Data Feeds: External research insights depend on RSS feeds from Hugging Face and MIT AI News; if unavailable, the system defaults to bundled static signals. Workforce data observations are synthetic defaults unless manually uploaded by the user.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☒ An architecture block diagram or system flow diagram is attached as Appendix A.

☒ Any use of official SDC sponsor resources has been declared in Section 5.

☒ The prototype described here is accessible to judges through the links provided in Section 1.

☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Cecilia Se

24 October 2026